

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	CSE , CSE(DS), AIDS
Programme:	B.E , B,TECH
Course Code & Course Name	CSA1708-ARTIFICIAL INTELLIGENCE
Academic Year / Batch	2 nd YEAR,2025 BATCH
Faculty Name	Dr.R.JEENA
Assignment Title	AI-BASED EARLY DETECTION OF ALZHEIMER'S DISEASE USING MEDICAL IMAGES
Date of Issue	02/09/26
Date of Submission	02/09/26
Maximum Marks	100
Course Outcome(s) – CO	CO5-AT3
Bloom's Taxonomy Level	Level 6 – Create: Design and develop an AI-based medical-image analysis system for early detection and risk classification of Alzheimer's disease.
SDG Mapping (SDG 1-17)	SDG 3 – Good Health and Well-Being: Supports early identification of Alzheimer's-related changes and improved healthcare decision support through AI-based medical-image analysis.
Industry / Societal Relevance	The project is relevant to the healthcare and medical-AI industry because it applies artificial intelligence and medical imaging to support early risk assessment of Alzheimer's disease. It can assist researchers and healthcare professionals by providing consistent image-based analysis, while supporting timely clinical evaluation and improving healthcare decision-making.

B. Assignment Problem / Challenge

Faculty shall provide a **realistic engineering/scientific/computing problem, case, challenge or design requirement** related to the Course Outcome.

The assignment should preferably require students to:

- Apply course concepts rather than reproduce theory.
- Identify and analyze the problem.
- Consider requirements and constraints.
- Develop or compare alternative solutions where appropriate.
- Use relevant modern engineering/computing/design tools.

AI-BASED EARLY DETECTION OF ALZHEIMER'S DISEASE USING MEDICAL IMAGES

B. Assignment Problem / Challenge

Alzheimer's disease (AD) is a progressive neurodegenerative condition in which subtle structural and functional brain changes can appear before severe clinical impairment. The engineering challenge is to design an artificial-intelligence system that analyzes medical images, identifies imaging patterns associated with early AD or mild cognitive impairment (MCI), and provides a reliable risk classification that can support—not replace—clinical assessment.

The proposed assignment focuses on a medical-image pipeline using brain MRI as the primary modality, with PET considered as an optional complementary modality. The system must address image quality, preprocessing, class imbalance, data leakage, model interpretability, and patient privacy. The 2024 Alzheimer's Association diagnostic framework recognizes anatomic MRI and FDG-PET as biomarkers of injury/dysfunction and amyloid PET and tau PET as core biological biomarkers, which motivates a multimodal-ready design.

The assignment requires application of image processing, pattern recognition, AI, evaluation metrics, visualization, and responsible-AI principles. The output is an academic prototype for early-risk screening and research; it is not a clinical diagnostic device.

- Apply medical-image processing and AI concepts rather than only describing them.
- Identify and analyze the early-detection problem and limitations of conventional image interpretation.
- Define functional, technical, privacy, safety, reliability, and ethical requirements.
- Develop a reproducible preprocessing and model-training workflow.
- Compare MRI and PET as potential inputs and justify the selected primary modality.
- Validate the system using sensitivity, specificity, F1-score, ROC-AUC and confusion-matrix analysis.

C. Problem Statement

Problem / Case / Design Challenge: Design an AI-based early detection framework that accepts standardized brain medical images, performs quality control and preprocessing, extracts clinically meaningful imaging patterns, and classifies a subject into a predefined research category such as cognitively normal (CN), mild cognitive impairment (MCI), or Alzheimer's disease (AD). The system should emphasize early-stage discrimination, explainability, reproducibility, and safe interpretation.

Expected output: a predicted class or risk score, confidence estimate, explanation/attention map highlighting influential brain regions, and a validation report containing quantitative performance measures.

The proposed framework is aligned with the type of longitudinal imaging data available through the Alzheimer's Disease Neuroimaging Initiative (ADNI), which provides MRI and PET data together with clinical and other biomarker information to approved researchers. ADNI's MRI archive includes DICOM images and quantitative summaries, while its PET component includes raw and processed PET images and regional amyloid/tau measures.

D. Requirements and Constraints

Requirement / Constraint	Proposed consideration
Functional	Accept a de-identified MRI study; perform quality checks; preprocess; infer CN/MCI/AD; generate an explanation map
Performance	Use sensitivity, specificity, precision, F1-score, ROC-AUC, confusion matrix and uncertainty intervals with appropriate statistical analysis
Technical	Use standardized orientation/resampling, intensity normalization, brain extraction where appropriate, and consistent preprocessing
Data	Prefer ADNI or another approved neuroimaging repository; use only authorized data.
Safety	System output is decision support only. A low-confidence result must be flagged for expert review.
Privacy	Do not expose names, identifiers, faces, or unnecessary metadata; follow de-identification
Reliability	Use independent testing, repeated runs or cross-validation, and leakage checks.

Requirement / Constraint	Proposed consideration
Ethics	Assess bias across age, sex, scanner/site and other relevant subgroups; avoid presenting research perfor
Resources	A GPU-enabled environment is preferred for 3D image models; storage and preprocessing time must be
Sustainability	Use efficient preprocessing and model architectures where possible and record computational requirem

E. Student Work

1. Problem Understanding and Formulation

Problem: Early AD-related changes may be subtle and distributed across brain regions. Manual interpretation is clinically valuable but can be time-consuming and may vary across observers. AI can learn image patterns from labeled datasets and provide consistent quantitative assistance.

Expected outcomes: the system should process a brain image consistently, classify the research category, provide an interpretable indication of influential regions, and quantify performance on unseen subjects.

Available information: MRI provides structural information such as brain anatomy and atrophy patterns. PET can provide functional or molecular information depending on tracer. ADNI combines imaging with clinical assessments and other biomarker data, making it suitable for research on longitudinal disease progression.

Assumptions: images are obtained under an approved protocol, labels are sufficiently reliable for the selected task, image preprocessing is reproducible, and subjects do not appear in more than one data split. The project assumes that the model is a screening/research aid and not a stand-alone diagnosis.

Constraints: limited sample size, class imbalance, scanner/site variation, motion and image-quality differences, possible demographic bias, computational cost, and the need to preserve patient privacy.

2. Application of Course Knowledge

The proposed system combines digital image processing, pattern recognition, supervised learning, model validation and explainable AI. Given an input image volume X , the model estimates class probabilities $P(y|X)$ for CN, MCI and AD. The predicted class is the category with the largest calibrated probability, while a confidence threshold can trigger human review.

Recommended model: a 3D Vision Transformer (ViT) or Swin Transformer can be used to learn spatial relationships from volumetric MRI patches. Transformer-based models can model long-range spatial dependencies and support attention-based visualization. If resources are limited, a 2D slice-based transformer can be evaluated as a baseline.

Preprocessing: DICOM/NIFTI conversion, orientation standardization, resampling to common voxel spacing, brain extraction where appropriate, intensity normalization, image-quality screening, and optional registration to a common template. Augmentation should be applied only to the training set.

Loss and optimization: cross-entropy is appropriate for a multi-class classification baseline. Class-weighted loss or balanced sampling can reduce class imbalance. AdamW with learning-rate scheduling is a practical optimizer choice.

Table 1. Medical imaging modalities relevant to Alzheimer's disease

The 2024 Alzheimer's Association workgroup places amyloid PET and tau PET among core imaging biomarkers, while anatomic MRI and FDG-PET are categorized as biomarkers of injury, dysfunction or degeneration. This distinction is important: the AI system should not treat all imaging modalities as interchangeable.

Modality	Primary information	Strength	Main limitation
Structural MRI	Brain anatomy and atrophy	Non-invasive; strong structural detail	Does not directly measure amyloid/t
FDG-PET	Glucose metabolism	Can capture metabolic dysfunction	Radiotracer exposure; cost and avail
Amyloid PET	Amyloid- β pathology	Direct biological biomarker	Cost, tracer availability and specializ
Tau PET	Tau pathology	Biological staging information	Higher complexity and availability c

3. Solution / Design / Methodology

The complete workflow is a reproducible pipeline: medical image → quality control → preprocessing → 3D patch embedding → transformer encoder → classification head → probability calibration → explanation map → validation report.

Stage	Operation	Output
1. Data acquisition	Obtain approved MRI/PET data and labels	De-identified image set + labels
2. Quality control	Check missing slices, motion/artifacts, orientation and metadata	QC-approved studies
3. Preprocessing	Standardize orientation, resample, brain extraction, normalize intensity	Model-ready volumes
4. Data split	Patient-level train/validation/test split	Leakage-controlled partitions
5. Augmentation	Small rotations, translations, intensity variation on training data only	Robust training samples
6. Model training	Transformer learns image-to-class mapping	Trained model
7. Calibration	Tune probability calibration on validation data	Reliable confidence scores
8. Explainability	Generate attention/occlusion maps and region summaries	Visual explanation
9. Testing	Evaluate on unseen subjects	Performance metrics
10. Validation	Compare baseline and subgroup analyses	Evidence for/against requirements

A key design decision is patient-level splitting. Multiple scans from the same subject must not be distributed across training and test sets because this can produce overly optimistic performance by allowing the model to recognize subject-specific patterns.

4. Use of Modern Tools

The following tools can be used to implement the proposed system. Exact software versions should be recorded in the final submission.

Tool	Purpose
Python	Data handling, preprocessing, model training and evaluation
PyTorch	Implementation of the transformer-based image classifier
NiBabel / SimpleITK	Reading and preprocessing NIFTI/DICOM-derived medical images
NumPy / Pandas	Numerical and tabular data processing
scikit-learn	Metrics, confusion matrix, ROC analysis and calibration utilities
Matplotlib	Graphs and result visualization
Jupyter / VS Code	Experiment development and documentation
GPU/CUDA	Acceleration of volumetric model training

Evidence of tool usage should include representative preprocessing outputs, a model-training log, validation plots, a confusion matrix, and an example explanation map. Any AI-assisted coding or writing used in the assignment should be acknowledged.

5. Results and Validation

Because no student-generated dataset or experiment output was supplied with this assignment, the values below are literature-derived benchmarks and proposed validation targets, not results claimed for the proposed prototype. A final implementation report should replace proposed targets with measurements obtained from the held-out test set.

Table 2. Proposed validation metrics

Metric	Formula / meaning	Why it matters
Accuracy	Correct predictions / all predictions	Overall classification performance
Sensitivity	TP / (TP + FN)	Important for reducing missed cases
Specificity	TN / (TN + FP)	Important for reducing false alarms
Precision	TP / (TP + FP)	Reliability of positive predictions
F1-score	2 × Precision × Recall / (Precision + Recall)	Balances precision and recall
ROC-AUC	Area under ROC curve	Measures discrimination across thresholds
Calibration	Agreement between confidence and observed frequency	Important for risk estimates

A 2024 meta-analysis of deep-learning MRI studies for AD/MCI reported pooled sensitivity of 0.84, specificity of 0.86 and AUROC of 0.92 across 18 studies. A later comparative meta-analysis reported pooled SROC-AUC values of 0.94 for structural MRI and 0.96 for 18F-FDG PET. These values are

context, not guaranteed performance for a new model.

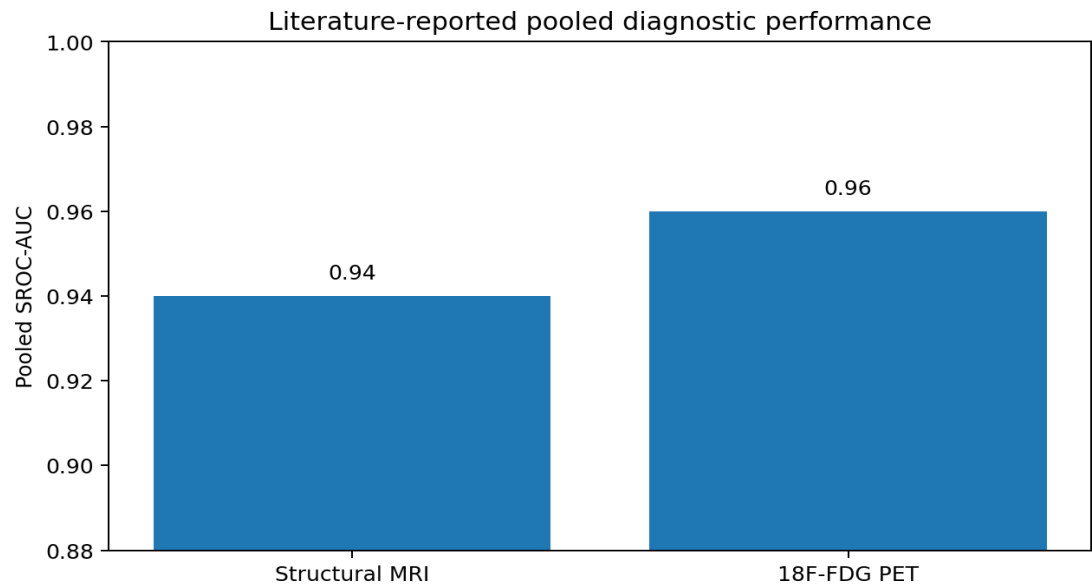


Figure 1. Selected literature-reported pooled SROC-AUC values. These are not experimental results from the proposed system and are not a direct head-to-head clinical trial.

Validation protocol: reserve an independent test set; report the confusion matrix; calculate sensitivity, specificity, precision, F1 and ROC-AUC; use bootstrapping for uncertainty intervals where feasible; inspect calibration; and perform subgroup checks by site/scanner and relevant demographic variables. If performance drops on an external dataset, report the limitation rather than hiding it.

6. Analysis and Engineering Decision

Why MRI is selected as the primary input: MRI is non-invasive and provides detailed anatomical information, making it practical for a first-stage academic prototype. ADNI also provides standardized MRI collections and quantitative outputs. PET can provide complementary metabolic or molecular information, but cost, availability, radiation exposure for PET, and multimodal registration make it more demanding for a basic implementation.

Why a transformer-based model is selected: a volumetric transformer can model relationships between distant brain regions and provides attention representations that can be inspected. The model is still subject to data limitations and should not be assumed to be clinically superior simply because it is more complex.

Trade-off: a larger model may improve representation capacity but increases GPU memory requirements, training time and overfitting risk. A smaller model is easier to reproduce but may miss subtle patterns. The recommended decision is to begin with a compact transformer baseline and increase complexity only after demonstrating a reproducible gain.

Decision factor	MRI-first transformer	Multimodal MRI + PET
Data availability	Higher for common research datasets	Lower and more restrictive
Cost / complexity	Lower	Higher
Structural information	Strong	Strong + complementary PET
Biological specificity	Lower than amyloid/tau PET	Potentially higher
Preprocessing complexity	Moderate	High: registration/fusion required
Recommended role	Primary academic prototype	Future extension / research comparison

The final decision is therefore an MRI-first, transformer-based, explainable research prototype, with PET kept as an optional extension. This balances scientific value, data availability, computational feasibility and responsible deployment.

7. Broader Considerations

- Sustainability: training volumetric models can require substantial GPU time. Efficient preprocessing, mixed precision and compact architectures can reduce computational demand.

- Society: earlier identification may support timely clinical evaluation and planning, but false positives or false negatives can create anxiety or delayed care if the system is misused.
- Safety: the system must display that its prediction is an aid and not a definitive diagnosis. Low-confidence or out-of-distribution cases should be escalated to a qualified clinician.
- Ethics: models can learn site-specific or demographic shortcuts. External validation and subgroup analysis are essential before clinical use.
- Privacy: medical images and associated clinical records are sensitive. Access control, de-identification and approved data-use agreements must be followed.
- Accessibility: results should be presented clearly and should not require users to interpret raw probability vectors without context.
- Professional responsibility: report limitations, dataset composition, preprocessing choices and failure cases rather than only the best metric.

8. Conclusion

This report proposes an AI-based early detection framework for Alzheimer's disease using medical images, with structural MRI as the primary modality and PET as a future complementary input. The workflow includes data quality control, standardized preprocessing, patient-level splitting, transformer-based image classification, probability calibration, explainability and rigorous validation. Literature evidence indicates that AI-assisted neuroimaging can achieve strong diagnostic discrimination, but performance varies across datasets, algorithms and validation strategies.

The proposed solution meets the assignment requirements by combining image-processing principles, AI methodology, modern tools, quantitative validation, engineering trade-offs and responsible-AI considerations. Its main limitation is that no student-generated experiment is included here; therefore, literature values are not presented as prototype results. A complete implementation should train and test the model on an approved dataset, perform external validation and document failure cases before any clinical interpretation.

9. Student Reflection

This assignment develops an understanding of how artificial intelligence can be applied to a real biomedical engineering problem rather than a purely synthetic classification task. The major learning outcome is that high model accuracy alone is not sufficient: data leakage, scanner differences, class imbalance, calibration, interpretability, privacy and clinical context can all affect usefulness.

An additional learning point is the difference between research classification and clinical diagnosis. A model may classify a curated dataset successfully while still failing on images from a different hospital or scanner. Therefore, external validation and transparent reporting are essential.

If additional time and resources were available, the project would be improved by adding an independent external cohort, comparing MRI-only with MRI+PET fusion, evaluating calibration, testing subgroup robustness and performing prospective-style validation. A clinician-in-the-loop review would also improve assessment of whether explanations are meaningful.

10. References

- [1] Alzheimer's Association Workgroup. "Revised criteria for diagnosis and staging of Alzheimer's disease." *Alzheimer's & Dementia*, 2024. doi:10.1002/alz.13859.
- [2] Wang, L.-x., Wang, Y.-z., Han, C.-g., Zhao, L., He, L., & Li, J. "Revolutionizing early Alzheimer's disease and mild cognitive impairment diagnosis: a deep learning MRI meta-analysis." *Arquivos de Neuro-Psiquiatria*, 82(8), 2024. doi:10.1055/s-0044-1788657.
- [3] Zia-Ur-Rehman, Awang, M. K., Ali, G., & Faheem, M. "Deep learning techniques for Alzheimer's disease detection in 3D imaging: A systematic review." *Health Science Reports*, 2024. doi:10.1002/hsr2.70025.
- [4] Jabli Mohamed Amine, M., & Moussa, M. "Multimodal Deep Learning Approaches for Early Detection of Alzheimer's Disease: A Comprehensive Systematic Review of Image Processing Techniques." *Current Alzheimer Research*, 2025. doi:10.2174/0115672050401817250721190509.
- [5] Rabinovici, G. D., et al. "Updated appropriate use criteria for amyloid and tau PET." *Alzheimer's & Dementia*, 2025. doi:10.1002/alz.14338.
- [6] Alzheimer's Disease Neuroimaging Initiative (ADNI). "MRI Image Data Sets / MRI and PET data documentation." Accessed 2026.

- [7] Comparative Diagnostic Accuracy of AI-Assisted 18F-FDG PET Versus Structural MRI in Alzheimer Disease: Systematic Review and Meta-Analysis. 2025. Pooled SROC-AUC values used in Figure 1: 0.94 for structural MRI and 0.96 for 18F-FDG PET.
- [8] National Institute on Aging / Alzheimer's Disease Neuroimaging Initiative data-use documentation, for dataset governance and access requirements.

Note on external sources and AI-assisted tools: This report was prepared using the provided assignment-format document as the structural template and publicly available scientific/technical sources for background and literature context. Any implementation should separately acknowledge the exact dataset, software versions, model checkpoints and AI-assisted tools actually used.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO-PO-Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	C05	1,2,3,4,5,8,9,11,12	L3/L4	10
Application of knowledge	C05	1,2,3,4,5,8,9,11,12	L3/L4	20
Solution / Design / Implementation	C05	1,2,3,4,5,8,9,11,12	L4/L5/L6	20
Modern tool usage	C05	1,2,3,4,5,8,9,11,12	L3/L4	10
Validation	C05	1,2,3,4,5,8,9,11,12	L4/L5	15
Analysis & justification	C05	1,2,3,4,5,8,9,11,12	L4/L5	15
Broader considerations	C05	1,2,3,4,5,8,9,11,12	L4/L5	5
Documentation & reflection	C05	1,2,3,4,5,8,9,11,12	L3/L4	5

Note: The CO/PO labels above are a proposed mapping for this report and should be adjusted to the official course outcomes supplied by the faculty. The provided template states that only genuinely assessed POs and Bloom's levels should be mapped.

H. Faculty Evaluation & Continuous Improvement

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%
Target CO Attainment	≥ 70% _____
Actual CO Attainment	90% _____

Faculty Analysis

Area	Faculty remarks
Areas in which students performed well:	Good knowledge of fundamental concepts..
Areas requiring improvement:	Needs more practice in applying concepts.
Corrective / Improvement Action:	Provide application-based exercises.
Action to be implemented in the next offering:	Include more real-time case studies.

Documents to be Maintained

- Assignment question/problem statement
- CO-PO and Bloom's mapping
- Assessment rubric
- Student submission
- Tool/simulation/experimental evidence, where applicable
- Evaluated script/report

- Marks and rubric-wise assessment
- CO attainment analysis
- Sample student work representing different performance levels
- Faculty analysis and continuous-improvement action